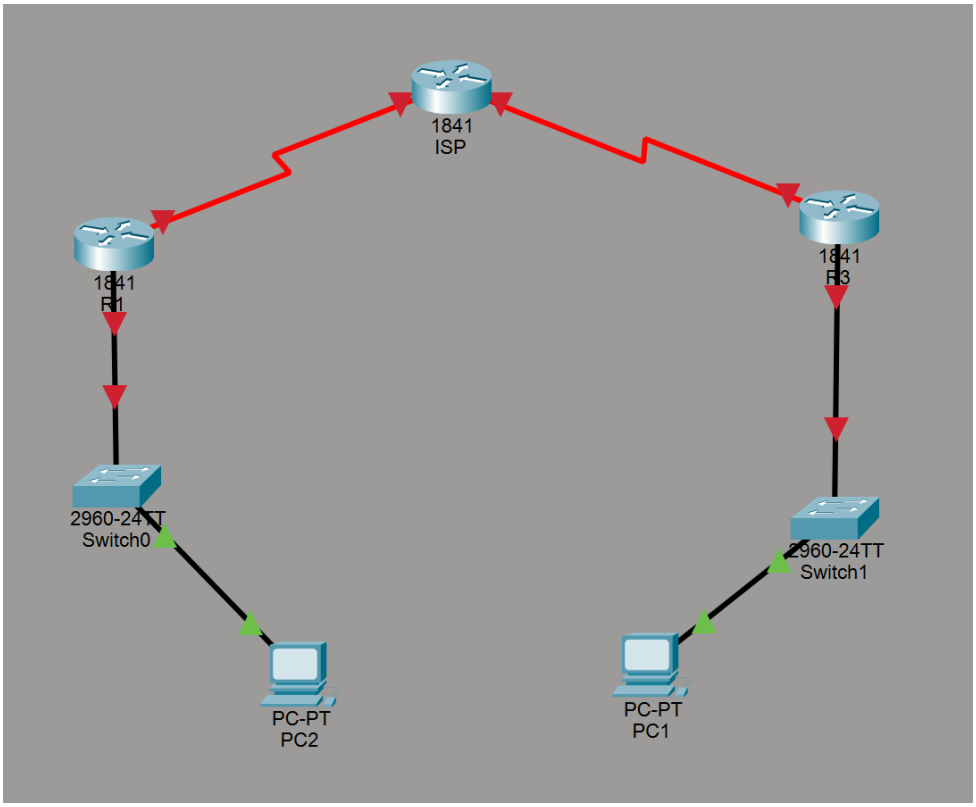

Your journal's **Practical-02 topology** uses:

- 3 Routers (R1, ISP, R3)
- 2 Switches (S1, S2)
- 2 PCs (PC-A, PC-C)

Harshika_SDN

1️⃣ Open Cisco Packet Tracer

Open Packet Tracer → **File → New**

2️⃣ Add Devices

Look at the **bottom device panel**.

Add Routers

Click:

Network Devices → Routers

Drag these to workspace:

3 × 1841 Router

Rename them:

R1
ISP
R3

Rename:

Right click router → Rename

Add Switches

Click:

Network Devices → Switches

Add:

2 × 2960-24TT

Rename:

S1
S2

Add PCs

Click:

End Devices

Add:

PC-PT (2)

Rename:

PC-A
PC-C

3️⃣ Arrange the Topology

Arrange like this:

```
PC-A   PC-C
  |      |
 S1     S2
  |      |
 R1     R3
   \   /
    ISP
```

4️⃣ Add Router Serial Ports

1841 routers **do not have serial ports by default**.

You must install them.

Do this for all routers

Click router → **Physical tab**

1. Turn **power OFF**
2. Drag module:

WIC-2T

into slot

3. Turn **power ON**

Now routers will have:

Serial0/0/0
Serial0/0/1

5️⃣ Connect Devices (VERY IMPORTANT)

Click bottom:

Connections (⚡ icon)

Connection 1 — PC-A to Switch S1

Cable:

Copper Straight Through

Ports:

PC-A FastEthernet0
→
S1 FastEthernet0/1

Connection 2 — PC-C to Switch S2

Cable:

Copper Straight Through

Ports:

PC-C FastEthernet0
→
S2 FastEthernet0/1

Connection 3 — S1 to R1

Cable:

Copper Straight Through

Ports:

S1 FastEthernet0/2
→
R1 FastEthernet0/1

Connection 4 — S2 to R3

Cable:

Copper Straight Through

Ports:

S2 FastEthernet0/2
→
R3 FastEthernet0/1

Connection 5 — R1 to ISP

Cable:

Serial DCE

Ports:

R1 Serial0/0/0
→
ISP Serial0/0/0

If Packet Tracer asks DCE/DTE → choose **DCE on ISP**.

Connection 6 — ISP to R3

Cable:

Serial DCE

Ports:

ISP Serial0/0/1
→
R3 Serial0/0/1

6️⃣ Final Topology

```
PC-A --- S1 --- R1 ---- ISP ---- R3 --- S2 --- PC-C
            |              |
        Serial Link     Serial Link
```

When connections work, the **green lights appear**.

7️⃣ Configure PC-A

Click **PC-A**

Desktop → IP Configuration

Enter:

IP Address: 192.168.10.3
Subnet Mask: 255.255.255.0
Gateway: 192.168.10.1

8️⃣ Configure PC-C

IP Address: 192.168.30.3
Subnet Mask: 255.255.255.0
Gateway: 192.168.30.1

9️⃣ Configure Router R1

Click **R1 → CLI**

Type:

```
enable
configure terminal
```

LAN interface

```
interface fastethernet0/1
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
```

Serial interface

```
interface serial0/0/0
ip address 10.1.1.1 255.255.255.252
no shutdown
exit
```

Loopback

```
interface loopback0
ip address 192.168.20.1 255.255.255.0
```

🔟 Configure ISP Router

```
enable
configure terminal

interface serial0/0/0
ip address 10.1.1.2 255.255.255.252
clock rate 64000
no shutdown

interface serial0/0/1
ip address 20.2.2.2 255.255.255.252
clock rate 64000
no shutdown

interface loopback0
ip address 209.165.200.225 255.255.255.224
```

1️⃣1️⃣ Configure Router R3

```
enable
configure terminal

interface fastethernet0/1
ip address 192.168.30.1 255.255.255.0
no shutdown

interface serial0/0/1
ip address 20.2.2.1 255.255.255.252
no shutdown

interface loopback0
ip address 192.168.40.1 255.255.255.0
```

1️⃣2️⃣ Configure RIP Routing

Do on **all routers**.

```
router rip
version 2
no auto-summary
```

On R1

network 192.168.10.0
network 10.1.1.0

On ISP

network 10.1.1.0
network 20.2.2.0

On R3

network 192.168.30.0
network 20.2.2.0

---

1 3 Test Connectivity

Go to **PC-A**

Desktop → Command Prompt

Run:

ping 192.168.30.3

If configuration correct → **Ping Successful**

---

1 4 Apply ACL

On **R3**

access-list 10 deny 192.168.10.3
access-list 10 permit any

Apply ACL:

interface fastethernet0/1
ip access-group 10 out

---

1 5 Test ACL

Now try again:

PC-A → ping 192.168.30.3

Ping should **fail**.

This proves ACL is working.

🚨 **Important Tip**

If lights are **red**, it usually means:

- forgot no shutdown
- wrong cable
- missing clock rate
- wrong IP